Full 17-Class Existing-Real Comparison

Calibrated Siamese+MLP 2-head vs Best Classical vs Best Deep

Scope. All three models are evaluated on the same 580-sample original real-mixture dataset from `mixtures_dataset.c` and all inference is scored against the full 17-class PT2-expanded label space.

Important note on the Siamese row. The calibrated Siamese thresholds come from the saved notebook recalibration experiment. To produce a full 17-class per-class comparison, those saved thresholds are applied here to the full 580-sample set. That makes this row a direct description of calibrated full-set behavior, not a held-out post-calibration estimate.

Table 1: Full 17-class summary on the original real-mixture dataset.

Model	Exact match	Micro-precision	Micro-recall	Micro-F1	Avg. returned chemi
Best classical	0.984	0.992	0.992	0.992	2.000
Best deep	0.952	0.974	0.974	0.974	2.000
Calibrated Siamese+MLP 2-head	0.283	0.564	0.946	0.707	3.352

Table 2: Per-class precision, recall, and F1 under full 17-class inference on the original real-mixture dataset.

Chemical	Support	Best classical			Best deep			Calibrated Siamese		
		P	R	F1	P	R	F1	P	R	F1
1,9-nonanedithiol	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1-dodecanethiol	243	1.000	1.000	1.000	1.000	0.942	0.970	0.859	0.979	0.915
1-undecanethiol	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
6-mercapto-1-hexanol	108	1.000	1.000	1.000	1.000	1.000	1.000	0.283	0.963	0.438
acetonitrile	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
benzene	193	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.974	0.987
benzenethiol	72	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.958	0.979
dichloromethane	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
diethylamine	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
dmmp	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
etoh	121	1.000	1.000	1.000	1.000	1.000	1.000	0.980	0.818	0.892
meoh	243	1.000	0.963	0.981	1.000	0.934	0.966	1.000	0.901	0.948
n,n-dimethylformamide	72	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n-hexane	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
pyridine	108	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
toluene	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
tris(2-ethylhexyl) phosphate	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

How the best classical model works. The classical system starts from a replicate-aware reference dictionary over the full 17-compound library. Each candidate compound contributes multiple measured pure spectra rather than one prototype. For one test spectrum, the solver enumerates all binary compound pairs, appends a nuisance baseline atom, and solves a nonnegative least-squares fit for each pair. The lowest-residual pair is the base prediction. The specific "best classical" row shown here is the strongest engineering variant in the repo: it keeps that exhaustive pairwise NNLS core, but adds a selective baseline-dependent fallback and a narrow near-tie family fallback to rescue the remaining dodecanethiol/methanol confusions. It always returns exactly two chemicals.

How the best deep model works. The promoted deep model is the similarity-supervised coefficient regressor. It trains on baseline-corrected synthetic binary mixtures built from the same 17-class reference library. The network takes a single spectrum as input and predicts a 17-dimensional nonnegative coefficient vector, one coefficient per library chemical. Training combines coefficient supervision with similarity-aware structure so chemically close spectra still have to separate in the output space. At inference time the model ranks all 17 coefficients and returns the top two chemicals. It does not use pair-specific thresholds or chemistry-specific fallback rules.

How the calibrated Siamese+MLP 2-head works. The original deep baseline uses two Siamese encoders: one for the Raman spectrum and one for its FFT representation. Their embeddings are concatenated and passed to a 17-way multilabel presence head that predicts one score per chemical. The calibrated variant then applies per-class thresholds learned on a calibration subset of the original real-mixture dataset. Unlike the classical and coefficient-regression models, it is not constrained to output two chemicals, so its calibrated thresholds can increase recall at the cost of extra false positives and a larger average predicted support.

Reading the comparison. The full 17-class table makes the operating difference visible. The classical and deep models concentrate almost all of their remaining error inside the thiol/methanol neighborhood while staying sparse, whereas the calibrated Siamese model overpredicts several classes that never appear in the original dataset at all, including `1,9-nonanedithiol`, `1-undecanethiol`, `diethylamine`, and `acetonitrile`. That is why its recall is high but its micro-precision and exact-match rate collapse under full 17-class scoring.